



Mathematics Part 1 — Cbse Annual Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	To factor the expression $50p^2 + 60pq + 18q^2$ using the identity $(a+b)^2 = a^2 + 2ab + b^2$, what must first be taken out as a common factor? (A) 2 (B) 3 (C) 5 (D) 6 Answer: Correct Option: (A) ► Why? A. The number 2 must be taken out as a common factor to yield $2[(5p)^2 + 2(5p)(3q) + (3q)^2]$, allowing the use of the identity. [Ch 4: Exploring Algebraic Identities]	1
2	Assertion (A): The distance between the points A (3, 4) and D (7, 1) is 5 units. Reason (R): The distance between two points is found by adding the difference in x-coordinates and the difference in y-coordinates. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. The assertion is true as the source shows $AD = \sqrt{4^2 + 3^2} = 5$ units. The reason is false because the distance is not found by simply adding the differences; rather, the source uses the formula $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$, which involves squaring the differences, summing them, and then taking the square root. [Ch 1: Orienting Yourself: The Use of Coordinates]	1



Q.NO.	QUESTIONS	MARKS
3	One diagonal of a rhombus is twice as long as the other. If the rhombus has area 128 cm^2, what is the length of the shorter diagonal? (A) $8\sqrt{2} \text{ cm}$ (B) 16 cm (C) 12 cm (D) $4\sqrt{2} \text{ cm}$ Answer: Correct Option: (A) ► Why? A. Let shorter diagonal = d_1, then longer diagonal = $2d_1$. Area = $\frac{1}{2} \times d_1 \times 2d_1 = d_1^2 = 128$, so $d_1 = \sqrt{128} = 8\sqrt{2} \text{ cm}$. [Ch 6: Measuring Space: Perimeter and Area]	1
4	Assertion (A): The side of a cube whose volume is $8n^3 - 60n^2m + 150nm^2 - 125m^3$ cubic units is $2n - 5m$ units. Reason (R): The expression $8n^3 - 60n^2m + 150nm^2 - 125m^3$ cannot be written in the form $(a-b)^3$ because the terms have negative coefficients. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. The assertion is true because the volume expression matches the expanded form of $(2n - 5m)^3$. The reason is false because the expression can indeed be written in the form $(a-b)^3$ as $(2n)^3 - 3(2n)^2(5m) + 3(2n)(5m)^2 - (5m)^3$. [Ch 4: Exploring Algebraic Identities]	1
5	A rational number in its lowest form has a denominator of $2^3 \times 5$. How many decimal places will its decimal expansion have? (A) 1 (B) 2 (C) 4 (D) 3 Answer: Correct Option: (D) ► Why? D. The number of decimal places in a terminating decimal is determined by the highest power of 2 or 5 in the denominator. Here, the highest power is 3, so it will have 3 decimal places. [Ch 3: The World of Numbers]	1



Q.NO.	QUESTIONS	MARKS
6	Assertion (A): $(x + y)(x^2 - xy + y^2) = x^3 + y^3$ is an identity. Reason (R): $(x - y)(x^2 + xy + y^2) = x^3 - y^3$ is an identity. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. Both the assertion and the reason are true as they are standard algebraic identities. However, the reason states a different identity and does not explain why the assertion is true. [Ch 4: Exploring Algebraic Identities]	1
7	For the relationship $^{\circ}\text{C} = a \cdot ^{\circ}\text{F} + b$, where ice melts at 0°C and 32°F, and water boils at 100°C and 212°F, what is the value of b? (A) $-160/9$ (B) $160/9$ (C) $-32/9$ (D) $32/9$ Answer: Correct Option: (A) ► Why? A. From $a = 5/9$ and $0 = 32a + b$, we get $b = -32 \cdot 5/9 = -160/9$. [Ch 2: Introduction to Linear Polynomials]	1
8	Which mathematician was the first to solve algebraic problems using geometry by interpreting them in terms of coordinates in the plane? (A) Brahmagupta (B) Ömar Khayyām (C) Al-Bīrūnī (D) Fermat Answer: Correct Option: (B) ► Why? B. Ömar Khayyām was the first mathematician to solve algebraic problems using geometry by interpreting them in terms of coordinates in the plane. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
9	To construct a line segment of length $\sqrt{2}$ on the number line, we measure $OA =$ (A) 1 unit (B) $\sqrt{2}$ units (C) 2 units (D) $\sqrt{3}$ units Answer: Correct Option: (B)	1



Q.NO.	QUESTIONS	MARKS
	► Why? B. By the Baudhāyana-Pythagoras Theorem, $OB^2 = OA^2 + AB^2 = 1^2 + 1^2 = 2$, so $OB = \sqrt{2}$ units. [Ch 3: The World of Numbers]	
10	On what date is Pi Approximation Day celebrated in India? (A) March 14 (B) March 22 (C) July 22 (D) July 14 Answer: Correct Option: (C) ► Why? C. 22 July (written '22-7' in India) is celebrated each year as Pi Approximation Day. [Ch 6: Measuring Space: Perimeter and Area]	1
11	Lines with equal slopes but different y-intercepts are always: (A) Perpendicular (B) Intersecting (C) Coincident (D) Parallel Answer: Correct Option: (D) ► Why? D. Lines with equal slopes but different y-intercepts are parallel to each other. [Ch 2: Introduction to Linear Polynomials]	1
12	In moving from point A(3, 4) to point D(7, 1), what is the distance covered along the x-axis? (A) 1 (B) 3 (C) 7 (D) 4 Answer: Correct Option: (D) ► Why? D. The distance moved along the x-axis is given by the x-coordinate of D minus the x-coordinate of A, which is $7 - 3 = 4$. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
13	For a square with side a, what is the ratio $P^2 : A$, where P is the perimeter and A is the area? (A) 16:1 (B) 4:1 (C) 8:1 (D) 2:1 Answer: Correct Option: (A)	1



Q.NO.	QUESTIONS	MARKS
	► Why? A. For a square, $P = 4a$ and $A = a^2$, so $P^2:A = 16a^2:a^2 = 16:1$. [Ch 6: Measuring Space: Perimeter and Area]	
14	Why does $\sqrt{-1}$ not exist on the real number line? (A) Because there is no real number that, when multiplied by itself, results in a negative number. (B) Because the square root operation is only defined for positive integers. (C) Because $\sqrt{-1}$ is a rational number that cannot be plotted. (D) Because negative numbers were not formalized until the 19th century. Answer: Correct Option: (A) ► Why? A. There is no real number that, when multiplied by itself, results in a negative number, as both $1 \times 1 = 1$ and $(-1) \times (-1) = 1$. [Ch 3: The World of Numbers]	1
15	What is the result of multiplying $(x + y + z)$ and $(x^2 + y^2 + z^2 - xy - xz - yz)$? (A) $x^3 + y^3 + z^3$ (B) $x^3 + y^3 + z^3 + 3xyz$ (C) $x^3 - y^3 - z^3 - 3xyz$ (D) $x^3 + y^3 + z^3 - 3xyz$ Answer: Correct Option: (D) ► Why? D. The product of these two expressions yields the identity $x^3 + y^3 + z^3 - 3xyz$. [Ch 4: Exploring Algebraic Identities]	1
16	Which of the following is a polynomial of degree 3 in the variable x with the coefficient of the x^2 term equal to -7? (A) $x^3 - 7x^2 + 2x + 1$ (B) $-7x^3 + 2x + 1$ (C) $3x^3 + 7x^2 - x + 5$ (D) $x^2 - 7x + 3$ Answer: Correct Option: (A) ► Why? A. The polynomial $x^3 - 7x^2 + 2x + 1$ has degree 3 (highest power of x is 3) and the coefficient of the x^2 term is -7. [Ch 2: Introduction to Linear Polynomials]	1
17	When simplifying the rational expression $(x^2 - 7x + 12)/(5x^2 + 5x - 100)$, assuming the denominator is not zero, what is the final simplified form? (A) $(x - 4)/(5(x + 5))$ (B) $(x - 3)/(5(x + 5))$ (C) $(x - 3)/(x + 5)$ (D) $(x - 4)/(x + 5)$	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (B) ► Why? B. The numerator factors to $(x - 3)(x - 4)$ and the denominator to $5(x - 4)(x + 5)$. Cancelling the common factor $(x - 4)$ gives $(x - 3)/(5(x + 5))$. [Ch 4: Exploring Algebraic Identities]	
18	A letter is picked at random from the word 'PROBABILITY'. What is the theoretical probability of picking the letter B? (A) 1/11 (B) 2/11 (C) 1/2 (D) 2/9 Answer: Correct Option: (B) ► Why? B. There are 2 'B's in the 11-letter word PROBABILITY, so the theoretical probability is 2/11. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	1
19	Assertion (A): The number of square tiles at Stage n of a growing pattern is given by $2n - 1$. Reason (R): The polynomial $2n - 1$ has degree 1, making it an example of a linear polynomial. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. Both A and R are true, but R is NOT the correct explanation of A. The formula for the number of squares at Stage n is indeed $2n - 1$, and this polynomial has degree 1, but the degree of the polynomial does not explain why the formula is $2n - 1$. [Ch 2: Introduction to Linear Polynomials]	1
20	Assertion (A): The equation $y = (9/5)(x - 273) + 32$ represents a linear relationship between the temperature in Kelvin and Fahrenheit. Reason (R): In the equation $y = ax + b$, if we change the values of b keeping the value of a fixed, the lines shift but remain parallel to the original line. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. The assertion is true because the equation $y = (9/5)(x - 273) + 32$ can be rearranged into the form $y = ax + b$, representing a linear relationship. The reason is true as changing 'b' shifts lines parallelly, but this property does not explain why the given equation is linear.	1



Q.NO.	QUESTIONS	MARKS
	[Ch 2: Introduction to Linear Polynomials]	
Section B Attempt 5 out of 7.		
21	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). Answer: relative frequency, 0.3 (or 15/50) ► Why? According to the provided context, the ratio of the number of times an event occurs to the total number of trials is called the relative frequency (e.g., relative frequency of rolling a 4 is $8/50 = 0.16$). Here, the number of students who like football is 15 out of a total of 50 students. Thus, the relative frequency is calculated as $15/50$, which simplifies to 0.3. ► Step-by-step: 1. Step 1: Identify the term for the ratio of observed occurrences to total trials, which is relative frequency. 2. Step 2: Calculate the relative frequency using the formula: Relative frequency = (Number of students who like football) / (Total number of students) = $15 / 50 = 0.3$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	2
	OR The height of water in a tank is given by $h(t) = 3 - 0.5t$. By what constant amount does the height decrease per month, and what does this represent? Answer: The height decreases by a fixed amount of 0.5 m for every month. Since the value of the height decreases by a constant amount over equal intervals, this represents linear decay. ► Why? A constant decrease in the dependent variable (height) for equal increments of the independent variable (time) defines linear decay. [Ch 2: Introduction to Linear Polynomials]	
22	How were the city streets of the Sindhu-Sarasvatī Civilisation constructed, and what distance apart were they? Answer: The city streets were constructed with striking precision in the North-South and East-West directions. They were laid out at uniform distances of about 10 metres apart, forming a coordinate system in practice. ► Why? This checks the historical context of grid-based thinking from the source, specifically the urban scale application in the Sindhu-Sarasvatī Civilisation. [Ch 1: Orienting Yourself: The Use of Coordinates]	2
	OR A die is rolled 50 times, and it lands on the number 4 exactly 8 times. Calculate the relative frequency of rolling a 4. Answer: Relative frequency is calculated as the number of times the event occurred divided by the total number of trials. Therefore, the relative frequency of rolling a 4 is $8/50 = 0.16$ or 16%.	



Q.NO.	QUESTIONS	MARKS
	► Why? This applies the formula for relative frequency (experimental probability) using the specific numbers provided, yielding the correct decimal and percentage. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	
23	Use a suitable identity to find the value of 43^2. Answer: We can write 43 as $40 + 3$. Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, we get $(40 + 3)^2 = 40^2 + 2 \times 40 \times 3 + 3^2$. Simplifying this gives $1600 + 240 + 9 = 1849$. ► Why? This tests the application of the $(a+b)^2$ identity for numerical calculation, mirroring Example 4 from the source text. [Ch 4: Exploring Algebraic Identities]	2
24	How did Archimedes express the area of a circle in terms of a right-angled triangle? Answer: Archimedes stated that the area of a circle equals the area of a right-angled triangle with sides equal to the radius and the circumference of the circle. This equals $\frac{1}{2} \times \text{circumference} \times \text{radius}$. ► Why? This tests the understanding of Archimedes' geometric interpretation of the area of a circle, connecting it to a triangle. [Ch 6: Measuring Space: Perimeter and Area]	2
25	Show that $\frac{2}{3}$ and $\frac{4}{6}$ are equal rational numbers using the equality rule. Answer: Two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$ are equal if $ad = bc$. For $\frac{2}{3}$ and $\frac{4}{6}$: $ad = 2 \times 6 = 12$ and $bc = 3 \times 4 = 12$. Since $ad = bc$, the two rational numbers are equal. ► Why? The source states the equality rule for rational numbers as $ad = bc$, which is applied here to verify the equivalence of the two given fractions. [Ch 3: The World of Numbers]	2
Section C Attempt 5 out of 7.		
26	Find 5 rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Given: Rational numbers 2/5 and 3/5. Formula: To find rational numbers between two numbers with the same denominator, select numerators between the given numerators. If insufficient integers exist, multiply numerator and denominator by a suitable integer. Substitution: Multiply both fractions by 6 to create space for 5 numbers: $(2 \times 6)/(5 \times 6) = 12/30$ $(3 \times 6)/(5 \times 6) = 18/30$ Calculation: The integers between 12 and 18 are 13, 14, 15, 16, and 17. Answer: The 5 rational numbers are 13/30, 14/30, 15/30, 16/30, and 17/30. ∴ This proves that rational numbers are dense, as we can always find more between any two given rationals. ► Why? By multiplying both fractions by 6, the denominators become 30 and the numerators become 12 and 18, leaving exactly 5 integer values (13 through 17) between them to form rational numbers. ► Step-by-step: 1. Step 1: Multiply the numerator and denominator of both fractions by a number (e.g., 6) to create a larger gap between numerators. 2. Step 2: Convert 2/5 to 12/30 and 3/5 to 18/30. 3. Step 3: Identify the 5 integers between 12 and 18, which are 13, 14, 15, 16, 17, and write them over 30. [Ch 3: The World of Numbers]	
	OR In the coordinate representation (x, y), what do the variables x and y specifically represent regarding the position of a point P in 2-D space? Answer: In the coordinate representation (x, y), the variables represent the perpendicular distances of the point P from the two coordinate axes. Specifically, x represents the perpendicular distance of point P from the y-axis, measured along the x-axis. y represents the perpendicular distance of point P from the x-axis, measured along the y-axis. Together, these two coordinates uniquely determine the exact physical location of the point in the 2-D plane. ► Why? The answer identifies x and y as perpendicular distances from the y-axis and x-axis respectively, as explicitly defined in the source text. ► Step-by-step: 1. Step 1: Identify that (x, y) represents the coordinates of a point P in 2-D space. 2. Step 2: State that x is the perpendicular distance from the y-axis, measured along the x-axis. 3. Step 3: State that y is the perpendicular distance from the x-axis, measured along the y-axis. [Ch 1: Orienting Yourself: The Use of Coordinates]	



Q.NO.	QUESTIONS	MARKS
27	Evaluate 65^2 using Śhrīdharāchārya's identity $a^2 = (a + b)(a - b) + b^2$. Answer: Given: 65^2 Formula: $a^2 = (a + b)(a - b) + b^2$ Substitution: Let $a = 65$ and $b = 5$ Calculation: $65^2 = (65 + 5)(65 - 5) + 5^2$ $= (70)(60) + 25$ $= 4200 + 25$ $= 4225$ ∴ The value of 65^2 is 4225. ► Why? Choosing a value for 'b' that rounds 'a' to the nearest multiple of 10 simplifies the multiplication into easily manageable numbers. ► Step-by-step: - Step 1: Choose a value for b that makes (a+b) and (a-b) easy to multiply. For 65, b = 5 works well as it gives 70 and 60. - Step 2: Substitute a = 65 and b = 5 into the identity to get $(65+5)(65-5) + 5^2$. - Step 3: Multiply 70 by 60 to get 4200, add 25, and arrive at 4225. [Ch 4: Exploring Algebraic Identities]	3
	OR In a village fair, there are 3 popular snacks available: Samosa, Pakora, and Bhaji. For drinks, villagers can choose either Chai or Lassi. (i) List the sample space of all possible snack and drink combinations a person could choose at the fair. (ii) List the event 'Selecting Samosa as a snack.' Answer: (i) $S = \{(\text{Samosa, Chai}), (\text{Samosa, Lassi}), (\text{Pakora, Chai}), (\text{Pakora, Lassi}), (\text{Bhaji, Chai}), (\text{Bhaji, Lassi})\}$ (ii) $E = \{(\text{Samosa, Chai}), (\text{Samosa, Lassi})\}$ ► Why? The sample space is formed by pairing each available snack with each available drink, representing all possible outcomes of the experiment. The event is a subset of the sample space that only includes outcomes satisfying the specific condition of choosing a Samosa. ► Step-by-step: - Step 1: Identify the options for snacks {Samosa, Pakora, Bhaji} and drinks {Chai, Lassi}. - Step 2: Form the sample space by pairing each snack with each drink. $S = \{(\text{Samosa, Chai}), (\text{Samosa, Lassi}), (\text{Pakora, Chai}), (\text{Pakora, Lassi}), (\text{Bhaji, Chai}), (\text{Bhaji, Lassi})\}$. - Step 3: Filter the sample space for the event 'Selecting Samosa as a snack'. $E = \{(\text{Samosa, Chai}), (\text{Samosa, Lassi})\}$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	



Q.NO.	QUESTIONS	MARKS
28	A straight line is represented by the equation $y = ax + b$. At what point does this line cut the y-axis? If a line is given by $y = 3x - 2$, what is its y-intercept and what does it signify? Answer: Y-axis Intersection: Any straight line written in the form $y = ax + b$ cuts the y-axis at the point $(0, b)$. The length b is referred to as the y-intercept of the line. Calculation: For the equation $y = 3x - 2$: Comparing with $y = ax + b$, we get $b = -2$. The line cuts the y-axis at $(0, -2)$. Significance: The y-intercept of -2 means that the line cuts the y-axis at a distance of 2 units from the origin in the negative direction. ► Why? In the slope-intercept form $y = ax + b$, setting $x=0$ yields $y=b$, meaning the line intersects the y-axis at $(0,b)$. A negative b indicates the intersection is below the origin. ► Step-by-step: - Step 1: Identify that the line $y = ax + b$ cuts the y-axis at the point $(0, b)$, where b is the y-intercept. - Step 2: For $y = 3x - 2$, identify $b = -2$, so it cuts the y-axis at $(0, -2)$. - Step 3: Explain that a y-intercept of -2 means the line cuts the y-axis 2 units from the origin in the negative direction. [Ch 2: Introduction to Linear Polynomials]	3
29	A point lies in Quadrant II and another point lies in Quadrant IV of the Cartesian plane. State the signs of the x-coordinate and y-coordinate for points in each of these two quadrants. Answer: The coordinate axes divide the Cartesian plane into four parts called quadrants, each with specific sign conventions for coordinates. In Quadrant II, the x-coordinate is negative and the y-coordinate is positive. For example, the point $Q(-5, 3)$ lies in this quadrant. In Quadrant IV, the x-coordinate is positive and the y-coordinate is negative. For example, the point $S(3, -5)$ belongs to this quadrant. These signs reflect the directions of measurement from the origin along the axes. ► Why? The answer details the sign conventions for Quadrants II and IV based on the text, using the provided examples of Q and S to illustrate the rules. ► Step-by-step: Step 1: Recall the sign conventions for the four quadrants of the Cartesian plane. Step 2: Identify that Quadrant II has a negative x-coordinate and positive y-coordinate. Step 3: Identify that Quadrant IV has a positive x-coordinate and negative y-coordinate. [Ch 1: Orienting Yourself: The Use of Coordinates]	3
30	When $d = 0$ in a 4-gon with sides a, b, c, d, show that the semi-perimeter of the 4-gon equals the semi-perimeter of the triangle with sides a, b, c.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Given: 4-gon with sides a, b, c, d where d = 0. Formula: Semi-perimeter of 4-gon = $\frac{1}{2}(a + b + c + d)$ Substitution: $s = \frac{1}{2}(a + b + c + 0)$ $s = \frac{1}{2}(a + b + c)$ $\therefore$ The semi-perimeter of the 4-gon equals $\frac{1}{2}(a + b + c)$, which is exactly the semi-perimeter of the triangle with sides a, b, c. This equivalence is essential for showing that Brahmagupta's formula reduces to Heron's formula when d = 0, since both formulas share the same value of s. ► Why? Setting $d = 0$ removes the fourth side from the semi-perimeter calculation, making $\frac{1}{2}(a+b+c+d)$ equal to $\frac{1}{2}(a+b+c)$, the standard semi-perimeter of the triangle. ► Step-by-step: - Step 1: Write the semi-perimeter of the 4-gon: $s = \frac{1}{2}(a + b + c + d)$ - Step 2: Substitute $d = 0$ - Step 3: Simplify to get $s = \frac{1}{2}(a + b + c)$ - Step 4: Conclude this equals the triangle's semi-perimeter [Ch 6: Measuring Space: Perimeter and Area]	
Section D Attempt 4 out of 6.		
31	The sum of three numbers x, y, and z is 6, their product is 8, and the sum of their squares is 14. Using the identity $(x + y + z)(x^2 + y^2 + z^2 - xy - xz - yz) = x^3 + y^3 + z^3 - 3xyz$, find the value of $x^3 + y^3 + z^3$.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Given: $x + y + z = 6$, $xyz = 8$, $x^2 + y^2 + z^2 = 14$ Formula 1: $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$ Formula 2: $(x + y + z)(x^2 + y^2 + z^2 - xy - xz - yz) = x^3 + y^3 + z^3 - 3xyz$ Step 1: Find $xy + xz + yz$ Substitution in Formula 1: $(6)^2 = 14 + 2(xy + xz + yz)$ $36 = 14 + 2(xy + xz + yz)$ $2(xy + xz + yz) = 36 - 14 = 22$ $xy + xz + yz = 11$ Step 2: Apply Formula 2 to find $x^3 + y^3 + z^3$ Substitution: $(6)(14 - 11) = x^3 + y^3 + z^3 - 3(8)$ $(6)(3) = x^3 + y^3 + z^3 - 24$ $18 = x^3 + y^3 + z^3 - 24$ Calculation: $x^3 + y^3 + z^3 = 18 + 24 = 42$ $\therefore x^3 + y^3 + z^3 = 42$ Explanation: The identity links the sum of cubes to the sum, sum of squares, product, and sum of pairwise products. We first derived $xy + xz + yz$ using the square of the sum identity, then substituted all known values into the cubic identity to obtain the result. ► Why? First find $xy + xz + yz$ using the square of sum identity, then substitute all known values into the cubic identity to solve for the sum of cubes. ► Step-by-step: - Step 1: Write down all given values and the two required identities. - Step 2: Use $(x+y+z)^2 = x^2+y^2+z^2+2(xy+xz+yz)$ to find $xy+xz+yz$. - Step 3: Substitute all known values into the identity $(x+y+z)(x^2+y^2+z^2-xy-xz-yz) = x^3+y^3+z^3-3xyz$. - Step 4: Solve the resulting equation for $x^3+y^3+z^3$. - Step 5: State the final answer with proper explanation. [Ch 4: Exploring Algebraic Identities]	



Q.NO.	QUESTIONS	MARKS
	OR Points A(3, 4) and D(7, 1) are plotted in the Cartesian plane. Using the Baudhāyana-Pythagoras Theorem, find the distance AD step by step. Then state the general formula for the distance between any two points (x_1, y_1) and (x_2, y_2). Verify that the formula works even when coordinates are negative, using A'(-3, 4) and D'(-7, 1). Answer: Given: A(3, 4) and D(7, 1) Step 1: Find the shift along the x-axis x-shift = x-coordinate of D - x-coordinate of A = 7 - 3 = 4 Step 2: Find the shift along the y-axis y-shift = y-coordinate of A - y-coordinate of D = 4 - 1 = 3 Step 3: Apply the Baudhāyana-Pythagoras Theorem AD = $\sqrt{(\text{x-shift})^2 + (\text{y-shift})^2} = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$ units General Formula: The distance between points (x_1, y_1) and (x_2, y_2) is: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ It makes no difference whether $(x_2 - x_1)$ and $(y_2 - y_1)$ are positive or negative, since we square these differences. Verification with negative coordinates: A'(-3, 4) and D'(-7, 1) x-shift = x-coordinate of A' - x-coordinate of D' = -3 - (-7) = 4 y-shift = y-coordinate of A' - y-coordinate of D' = 4 - 1 = 3 A'D' = $\sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$ units ∴ The distance remains the same, confirming that reflection preserves lengths and the formula works regardless of the signs of the coordinates. ► Why? The distance between two points is found by computing the horizontal and vertical shifts between them and applying the Baudhāyana-Pythagoras Theorem; squaring the differences ensures the formula works for negative coordinates too. ► Step-by-step: 1. Step 1: Calculate the shift along the x-axis: $x_2 - x_1$. 2. Step 2: Calculate the shift along the y-axis: $y_1 - y_2$. 3. Step 3: Apply the Baudhāyana-Pythagoras Theorem: distance = $\sqrt{(x\text{-shift})^2 + (y\text{-shift})^2}$. 4. Step 4: State the general formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and explain why signs do not matter. 5. Step 5: Verify with negative coordinates A'(-3, 4) and D'(-7, 1) to show the result is the same.	



Q.NO.	QUESTIONS	MARKS										
	[Ch 1: Orienting Yourself: The Use of Coordinates]											
32	A student has ₹500 in her savings bank account. She gets ₹150 every month as pocket money. Construct a table of values showing the total amount she will have at the end of the first 4 months. Find a linear expression to represent the amount she will have in the nth month. Explain why this represents linear growth. Answer: Given: Initial amount = ₹500. Monthly pocket money = ₹150. Calculation of Monthly Amounts: End of month 1: 500 + 1 × 150 = ₹650 End of month 2: 500 + 2 × 150 = ₹800 End of month 3: 500 + 3 × 150 = ₹950 End of month 4: 500 + 4 × 150 = ₹1100 Table of Values: <table><tr><td>Month (n)</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Amount (₹)</td><td>650</td><td>800</td><td>950</td><td>1100</td></tr></table> Linear Expression: The amount at the end of the nth month will be: ₹(500 + 150n) Explanation of Linear Growth: As the value of n increases by one month, the value of the total amount increases by a fixed amount of ₹150. Linear growth describes a linear pattern where a quantity increases by a constant amount over equal intervals. Therefore, this represents linear growth.	Month (n)	1	2	3	4	Amount (₹)	650	800	950	1100	5
Month (n)	1	2	3	4								
Amount (₹)	650	800	950	1100								



Q.NO.	QUESTIONS	MARKS
	► Why? The total amount increases by a constant value of ₹150 for each additional month, forming an arithmetic progression. This constant increase over equal intervals characterizes a linear growth pattern. ► Step-by-step: - Step 1: Calculate the total amount at the end of the first 4 months by adding the initial amount to (number of months $\times$ 150). - Step 2: Construct a table of values showing the month number and the corresponding amount. - Step 3: Generalize the pattern to find a linear expression for the amount in the n^{th} month. - Step 4: Explain that the expression represents linear growth because the amount increases by a constant value (₹150) over equal intervals (each month). [Ch 2: Introduction to Linear Polynomials]	
	OR A basket contains 4 red balls and 5 blue balls. One ball is drawn and laid aside, and a second ball is drawn. Draw a tree diagram to represent the possible outcomes and probabilities. Use the tree diagram to calculate the probability of drawing a red ball and then a blue ball, and the probability of drawing 2 blue balls.	



Q.NO.	QUESTIONS	MARKS
	Answer: Given: A basket contains 4 red balls and 5 blue balls. Total balls = 9. One ball is drawn, not replaced, and a second ball is drawn. Step 1: Probabilities for the First Draw Probability of drawing a red ball first: $P(R_1) = 4/9$ Probability of drawing a blue ball first: $P(B_1) = 5/9$ Step 2: Probabilities for the Second Draw (without replacement) If the first ball was red, 8 balls remain (3 red, 5 blue): Probability of drawing a red ball second: $P(R_2 \text{ mid } R_1) = 3/8$ Probability of drawing a blue ball second: $P(B_2 \text{ mid } R_1) = 5/8$ If the first ball was blue, 8 balls remain (4 red, 4 blue): Probability of drawing a red ball second: $P(R_2 \text{ mid } B_1) = 4/8 = 1/2$ Probability of drawing a blue ball second: $P(B_2 \text{ mid } B_1) = 4/8 = 1/2$ Step 3: Tree Diagram Description Draw a single starting point. Branch it into two paths: "Red (4/9)" and "Blue (5/9)". From the "Red" branch, draw two new branches: "Red (3/8)" and "Blue (5/8)". From the "Blue" branch, draw two new branches: "Red (4/8)" and "Blue (4/8)". Step 4: Probability of Red then Blue $P(\text{Red then Blue}) = P(R_1) \times P(B_2 \text{ mid } R_1) = 4/9 \times 5/8 = 20/72 = 5/18$ Step 5: Probability of 2 Blue Balls	



Q.NO.	QUESTIONS	MARKS
	$P(2 \text{ Blue Balls}) = P(B_1) \times P(B_2 \text{ mid } B_1) = 5/9 \times 4/8 = 20/72 = 5/18$ $\therefore$ The probability of drawing a red ball and then a blue ball is 5/18, and the probability of drawing 2 blue balls is 5/18. ► Why? This problem involves dependent events because the first ball is not replaced, altering the probabilities for the second draw. A tree diagram visually maps these sequential outcomes and their probabilities. ► Step-by-step: - Step 1: Calculate the probabilities for the first draw (Red = 4/9, Blue = 5/9). - Step 2: Calculate the conditional probabilities for the second draw based on the first outcome. - Step 3: Describe the tree diagram structure with branches for each outcome and its probability. - Step 4: Multiply probabilities along the 'Red then Blue' path to find $P(\text{Red then Blue})$. - Step 5: Multiply probabilities along the 'Blue then Blue' path to find $P(2 \text{ Blue Balls})$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	
33	Find the perimeter of a sector of a circle of radius 14 cm and sector angle 75°. Explain the steps involved in calculating the arc length and the total perimeter of the sector. (Use $22/7$ for π)	5



Q.NO.	QUESTIONS	MARKS
	Answer: Given: Radius of sector = 14 cm, Sector angle = 75°, $\pi \approx 22/7$ Formula: Length of arc = $2\pi r \times \theta^\circ/360^\circ$ Perimeter of sector = Length of arc + $2 \times$ radius Substitution: Length of arc = $2 \times 22/7 \times 14 \times 75^\circ/360^\circ$ Perimeter of sector = Length of arc + 2×14 Calculation: Length of arc = $2 \times 22/7 \times 14 \times 5/24$ = $2 \times 22 \times 2 \times 5/24$ = $88 \times 5/24$ = $440/24 = 55/3 = 18.33$ cm Perimeter of sector = $18.33 + 28 = 46.33$ cm $\therefore$ The perimeter of the sector is 46.33 cm. A sector is a portion of a circle enclosed by two radii and an arc. The perimeter of a sector includes the curved arc length as well as the two straight radii. The length of an arc depends on the angle it subtends at the centre. The formula $2\pi r \times \theta^\circ/360^\circ$ calculates the fraction of the total circumference based on this central angle. For example, a semicircle subtends 180°, making its length $2\pi r \times 180^\circ/360^\circ = \pi r$. Similarly, a quarter circle subtends 90°, giving an arc length of $(\pi r)/2$. To find the total perimeter, we add the lengths of the two straight portions (the radii) to the calculated arc length. Diagram description: Draw a circle with centre O. Mark two points A and B on the circumference. Join OA and OB. Shade the region AOB, which is the sector. Label the radius as 14 cm and the angle AOB as 75°.	



Q.NO.	QUESTIONS	MARKS
	► Why? The perimeter of a sector is the sum of its arc length and the two radii. The arc length is found by taking the proportional fraction of the total circumference based on the central angle. ► Step-by-step: - Step 1: Define a sector and state the formula for the length of an arc: $2\pi r \times \theta^\circ/360^\circ$. - Step 2: Substitute the given values ($r = 14 \text{ cm}$, $\theta = 75^\circ$, $\pi \approx 22/7$) into the arc length formula and calculate the arc length. - Step 3: State that the perimeter of the sector is the sum of the arc length and the lengths of the two radii. - Step 4: Calculate the total perimeter by adding the arc length to $2 \times 14 \text{ cm}$. - Step 5: Describe the diagram of the sector showing the centre, the two radii, the arc, and the angle of 75°. [Ch 6: Measuring Space: Perimeter and Area]	
34	Prove that $\sqrt{2}$ is an irrational number using the proof by contradiction method. Explain each logical step clearly and state why the initial assumption leads to a contradiction.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Proof by Contradiction is a technique where we assume the opposite of what we want to prove, and show that this assumption leads to a logical disaster. Step 1 — Assumption: Assume $\sqrt{2}$ is a rational number. Therefore, it can be written as a fraction p/q, $q \neq 0$, in its simplest form, meaning p and q are co-prime (they share no common factors other than 1). $\sqrt{2} = p/q$ Step 2: Square both sides: $2 = p^2/q^2$ Step 3: Multiply both sides by q^2: $2q^2 = p^2$ Step 4 — Deduction for p: Since p^2 equals 2 times an integer, p^2 must be even. If the square of a number is even, the number itself must be even. Therefore, p is even. Let $p = 2k$ (where k is an integer). Step 5: Substitute $p = 2k$ into the equation from Step 3: $2q^2 = (2k)^2$ $2q^2 = 4k^2$ Step 6: Divide both sides by 2: $q^2 = 2k^2$ Step 7 — Deduction for q: Now q^2 equals 2 times an integer k^2, so q^2 is even, and therefore q must also be even. Step 8 — The Contradiction: We deduced that both p and q are even, meaning they share a common factor of 2. However, in Step 1, we stated p/q was in simplest form with no common factors! This is a logical contradiction. Since our logical steps are flawless, our initial assumption must be wrong. $\therefore \sqrt{2}$ is irrational. ► Why? The proof assumes $\sqrt{2}$ is rational, expresses it as p/q in lowest terms, and shows both p and q must be even — contradicting the assumption that they are co-prime. ► Step-by-step: 1. Step 1: Assume $\sqrt{2}$ is rational and write it as p/q in simplest form (co-prime). 2. Step 2: Square both sides to get $2 = p^2/q^2$, then rearrange to $2q^2 = p^2$. 3. Step 3: Deduce that p^2 is even, so p is even; write $p = 2k$. 4. Step 4: Substitute $p = 2k$ back to get $q^2 = 2k^2$, deducing q is also even. 5. Step 5: Conclude that both p and q being even contradicts the co-prime assumption, so $\sqrt{2}$ is irrational.	



Q.NO.	QUESTIONS	MARKS
	[Ch 3: The World of Numbers]	
Section E Attempt 3 out of 4.		
35	Case Study: Using Identities for Quick Calculations A mathematics club is exploring how algebraic identities simplify calculations. Rahul computes 43^2 by writing it as $(40 + 3)^2$ and using the identity $(a + b)^2 = a^2 + 2ab + b^2$. Priya calculates 104×96 by expressing it as $(100 + 4)(100 - 4)$ and applying $(x + y)(x - y) = x^2 - y^2$. The teacher explains that identities are equations true for all values of variables, unlike equations like $x^2 - 1 = 24$ which hold only for specific values such as $x = 5$ or $x = -5$. (i) Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, find the value of 43^2 by writing it as $(40 + 3)^2$. (ii) Using the identity $(x + y)(x - y) = x^2 - y^2$, calculate 104×96. (iii) The expression $x^2 - 1 = 24$ is true only for $x = 5$ or $x = -5$. Is this expression an identity? Justify your answer. (iv) Using the expansion of $(a + b)^2$, determine when $(a + b)^2$ will be equal to $a^2 + b^2$. Answer: (i) $(40 + 3)^2 = 40^2 + 2 \times 40 \times 3 + 3^2 = 1600 + 240 + 9 = 1849$ (ii) $104 \times 96 = (100 + 4)(100 - 4) = 100^2 - 4^2 = 10000 - 16 = 9984$ (iii) No, it is not an identity. An identity is an equation that is true for all values of the variables. The expression $x^2 - 1 = 24$ holds only for $x = 5$ and $x = -5$, not for all values of x. (iv) $(a + b)^2 = a^2 + b^2$ when $2ab = 0$, which happens when at least one of a or b is zero. ► Why? i) $(40 + 3)^2 = 40^2 + 2 \times 40 \times 3 + 3^2 = 1600 + 240 + 9 = 1849$ ii) $104 \times 96 = (100 + 4)(100 - 4) = 100^2 - 4^2 = 10000 - 16 = 9984$ iii) No, it is not an identity. An identity is an equation that is true for all values of the variables. The expression $x^2 - 1 = 24$ holds only for $x = 5$ and $x = -5$, not for all values of x. iv) $(a + b)^2 = a^2 + b^2$ when $2ab = 0$, which happens when at least one of a or b is zero. [Ch 4: Exploring Algebraic Identities]	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: City Map Coordinates A student is creating a 2-D map of a city center, inspired by how early Arabic maps used the Ujjayinī meridian as a zero-longitude reference. On the map, the main intersection is chosen as the point of intersection of the axes, called the origin. The horizontal road is the x-axis and the vertical road is the y-axis. A cafe is located 3 units to the right and 2 units upwards from the origin. A library is located 4 units to the left and 5 units upwards from the origin. A park is located 2 units to the left and 3 units downwards from the origin. The student marks these locations using coordinates. (i) What are the coordinates of the origin? (ii) The park is located 2 units to the left and 3 units downwards from the origin. In which quadrant does the park lie? (iii) The cafe is at (3, 2). A shop is located at (7, 2). Find the distance between the cafe and the shop. (iv) A bookstore is located on the y-axis, 6 units upwards from the origin, and a florist is located on the x-axis, 6 units to the right of the origin. Are the coordinates of the bookstore and the florist the same? Explain. Answer: (i) (0, 0) (ii) The third quadrant (iii) The distance between points (x1, y) and (x2, y) is $x_2 - x_1$. Here, distance = $7 - 3 = 4$ units. (iv) No, the bookstore is at (0, 6) and the florist is at (6, 0). Since $0 \neq 6$, $(0, 6) \neq (6, 0)$. ► Why? i) (0, 0) ii) The third quadrant iii) The distance between points (x1, y) and (x2, y) is $x_2 - x_1$. Here, distance = $7 - 3 = 4$ units. iv) No, the bookstore is at (0, 6) and the florist is at (6, 0). Since $0 \neq 6$, $(0, 6) \neq (6, 0)$. [Ch 1: Orienting Yourself: The Use of Coordinates]	
36	Case Study: Mobile Network Plans and Linear Relationships Two mobile networks offer different monthly plans. Network A's monthly charge is given by the linear relationship $y = 2x + 10$, where x is the data used in GB and y is the total cost in ₹. Network B's plan is given by $y = 2x + 25$. Both networks have graphs that are straight lines. Because their equations have the same coefficient of x but different constant terms, their graphs are parallel lines. A third network offers a plan given by $y = 5x$. The constant term in each equation represents the y-intercept, which is the distance from the origin where the line cuts the y-axis. (i) What is the slope of the line representing Network A's plan? (ii) The graphs of Network A and Network B are parallel lines. What is the relationship between their slopes? (iii) The constant term in the equation $y = 2x + 25$ is the y-intercept. What does the y-intercept represent geometrically? (iv) The line representing the third network's plan, $y = 5x$, has no constant term ($b = 0$). Through which specific point does this line pass?	5



Q.NO.	QUESTIONS	MARKS
	Answer: (i) The slope of the line representing Network A's plan is 2. (ii) Their slopes are equal (or the same), as parallel lines have a fixed slope. (iii) The y-intercept is the distance from the origin where the line cuts the y-axis. (iv) The line passes through the origin. ► Why? i) The slope of the line representing Network A's plan is 2. ii) Their slopes are equal (or the same), as parallel lines have a fixed slope. iii) The y-intercept is the distance from the origin where the line cuts the y-axis. iv) The line passes through the origin. [Ch 2: Introduction to Linear Polynomials]	
37	Case Study: Room Layout with Coordinates Reiaan is planning his room layout on a coordinate plane. The room door extends from $D_1(9.5, 0)$ to $R_1(11.5, 0)$. The bathroom door extends from $B_1(0, 1.5)$ to $B_2(0, 4)$. A rectangular study table has three of its feet at the points $(8, 9)$, $(11, 9)$ and $(11, 7)$. Reiaan wants to verify the dimensions of each item and check whether the bathroom door is wider or narrower than the room door. He also needs to find the position of the fourth foot of the table to complete the layout. (i) What is the width of the room door? (ii) What are the coordinates of the fourth foot of the rectangular study table? (iii) Find the width of the bathroom door. Is the bathroom door narrower or wider than the room door? (iv) Both B_1 and B_2 have x-coordinate equal to 0. What does this indicate about the position of the bathroom door with respect to the y-axis? Answer: (i) Width = $11.5 - 9.5 = 2$ units (ii) $(8, 7)$. Since the table is rectangular, the fourth foot must have the same x-coordinate as $(8, 9)$ and the same y-coordinate as $(11, 7)$. (iii) Width of bathroom door = $4 - 1.5 = 2.5$ units. The bathroom door is wider than the room door ($2.5 > 2$). (iv) The bathroom door lies along the y-axis, since both its endpoints have x-coordinate 0, meaning their perpendicular distance from the y-axis is zero. ► Why? i) Width = $11.5 - 9.5 = 2$ units ii) $(8, 7)$. Since the table is rectangular, the fourth foot must have the same x-coordinate as $(8, 9)$ and the same y-coordinate as $(11, 7)$. iii) Width of bathroom door = $4 - 1.5 = 2.5$ units. The bathroom door is wider than the room door ($2.5 > 2$). iv) The bathroom door lies along the y-axis, since both its endpoints have x-coordinate 0, meaning their perpendicular distance from the y-axis is zero. [Ch 1: Orienting Yourself: The Use of Coordinates]	5

